

4. Project Planning Phase

Project Module: Visitor Management System: -

Phase Description:

Implementation roadmap, development sequencing, workflow planning, validation strategy, and testing preparation for the Visitor Management System.

This document serves as the formal milestone artifact for the ServiceNow development workflow lifecycle.

Work Structure:

Application Scope Creation:

Created a dedicated scoped application named:

Visitor Management System:

This ensured that all tables, scripts, flows, UI actions, and notifications remained isolated from the global ServiceNow environment.

Component Sequencing:

The project was planned and developed in the following order:

- Create Custom Application
- Create Visitor Table
- Configure Table Fields
- Enable Auto Numbering
- Configure Choice Fields
- Configure Reference Fields
- Develop Client Script Validation
- Build Flow Designer Automation
- Configure Email Notifications
- Create Check in UI Action
- Perform Testing & Validation

- Prepare Documentation

Form Design Planning:

The Visitor Registration Form was designed using a structured layout to ensure ease of use and logical data entry.

Field arrangement sequence:

- Visitor Name
- Employee Name
- Phone Number
- Email
- Company Name
- Visit Date
- Purpose of Visit
- Status
- Checked In Time

The layout was planned to improve usability and minimize user input errors.

Workflow Planning: -

Visitor Registration Process:

Planned workflow:

Visitor Registration

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Phone Number Validation

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Record Creation

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Automatic Approval

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Email Notification

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Visitor Check In

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Status Update

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Check-In Time Recording

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Process Complete

Automated Approval Process:

The Flow Designer workflow was planned to automatically process visitor records after creation.

Actions included:

- Trigger on Record Creation
- Update Visitor Status
- Send Approval Email
- Record Workflow Completion

Notification Planning:

The system was designed to automatically notify visitors when their registration request was approved.

Notification objectives:

- Improve communication
- Reduce manual follow-up
- Provide real-time updates

UI Action Planning: -

Check In Functionality:

A custom UI Action was planned for security and reception staff.

Expected actions:

- Change visitor status to Checked In
- Capture current date and time
- Display success confirmation message
- Update visitor records automatically

Security & Operational Benefits:

- Faster visitor processing
- Accurate attendance records
- Reduced manual data entry
- Improved visitor tracking

Testing Plan & Checklist: -

Form Validation Testing:

Verify:

- Visitor Name accepts valid input.
- Employee Name references valid users.
- Email field stores correct values.
- Visit Date accepts valid dates.

Phone Number Validation Testing:

Verify:

- Numbers less than 10 digits are rejected.
- Numbers greater than 10 digits are rejected.
- Valid 10-digit numbers are accepted.

Expected Message:

"Phone Number must contain exactly 10 digits."

Auto Number Testing:

Verify:

- Each visitor record receives a unique number.
- Numbers are generated sequentially.

Example:

VIS0001001

VIS0001002

VIS0001003

Flow Designer Testing:

Verify:

- Flow triggers when a visitor record is created.
- Status automatically changes from Requested to Approved.
- Email notification is generated successfully.

UI Action Testing:

Verify:

- Check In button appears only for approved visitors.
- Clicking Check In changes status to Checked In.
- Check-In Time is recorded automatically.
- Success message is displayed.

Email Notification Testing:

Verify:

- Email is sent after visitor approval.
- Correct visitor details are included.
- Subject line displays properly.

Risk Assessment & Mitigation:

Risk	Mitigation Strategy
Invalid phone numbers	Implement Client Script validation
Missing visitor information	Use mandatory field configuration
Manual approval delays	Automate approval using Flow Designer
Inaccurate check-in tracking	Use UI Action with automatic timestamp
Communication delays	Configure automated email notifications

Planning Outcome:

The project planning phase successfully established a structured development roadmap for implementing the Visitor Management System. The planned architecture ensured proper sequencing of development activities, minimized implementation risks, and enabled successful automation of visitor registration, approval, notification, and check-in processes.